

Predicting Falcon 9 First Stage Landings



IBM Data Science Professional Certificate – Applied Data Science
Capstone

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Github Link:

- <https://github.com/M-Denton369/IBM-Data-Science-Capstone-SpaceX>

OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

EXECUTIVE SUMMARY



Point 1: Project Overview & Core Objective

- The primary goal of this project was to analyze rocket launch data to predict whether the Falcon 9 first stage would successfully land, and to identify the cost and operational factors impacting landing success.

Point 2: Key Methodology & Tools

- Gathered and wrangled data using SpaceX APIs, web scraping, and SQLite/PostgreSQL databases via SQL queries.
- Conducted Exploratory Data Analysis using Pandas, Matplotlib, Seaborn, and built interactive geographic visualizations using Folium maps.
- Developed interactive dashboards with Plotly Dash and trained multiple machine learning classification models to predict landing outcomes.

Point 3: Major Analytical Findings

- EDA revealed that launch success rates increased significantly over time (learning curve), and payload mass, booster version, and launch site location impacted landing viability.

Point 4: Best-Performing Predictive Model Results

- After testing multiple algorithms (Logistic Regression, SVM, Decision Trees, and KNN), the Support Vector Machine (SVM) / Decision Tree model achieved the highest predictive accuracy on the test data.

Point 5: Final Strategic Conclusion & Recommendation

- The analysis proved that past operational data can reliably predict landing outcomes, providing actionable cost-saving insights for future aerospace logistics planning.

INTRODUCTION



- Point 1: Background and Context
- The commercial aerospace industry is shifting toward reusable rocket technology to reduce the cost of access to space significantly. SpaceX advertises Falcon 9 launches at around \$62 million, while other providers charge around of \$165 million
- Point 2: Core Problem Statement
- Determining factors that lead to successful booster landings is crucial for operational planning. Most of the savings come from recovering and reusing the first stage booster
- Point 3: Objectives
- To use data science, SQL, and machine learning to analyze past launch patterns and build a predictive model for first stage landing outcomes.
- Point 4: Overview
- Data acquisition, cleaning, and relational database storage using Python and SQL. Also analysis including statistical visualizations, interactive web mapping, custom dashboards, and predictive algorithms.



METHODOLOGY



- Point 1: Data collection: SpaceX REST API and web scraping of Wikipedia launch tables
- Point 2: Data Wrangling: Filtered to Falcon 9, filled missing payloads, created the Class label
- Point 3: EDA: Visualization with Matplotlib/Seaborn and queries with SQL
- Point 4: Interactive Visuals: Folium maps and a Plotly Dash dashboard
- Point 5: Predictive Analysis: Built, tuned and evaluated 4 classification models

Data collection and data wrangling methodology slides (1 pt)

- **SpaceX REST API**

- GET requests to `api.spacexdata.com/v4/launches/past`
- Follow-up lookups by ID for rocket, launchpad, payload and core details
- JSON normalized into a pandas DataFrame
- Returned booster version, payload mass, orbit, site, landing outcome, grid fins, legs, reuse and coordinates

- **Web scraping (Wikipedia)**

- Requested the 'List of Falcon 9 and Falcon Heavy launches' page
- Parsed the HTML launch tables with BeautifulSoup
- Extracted column names from table headers, then each launch row
- Captured flight numbers, date, boosters, sites, payloads, orbits, customer and outcomes

Data Collection – SpaceX API & Webscraping

- 90 Falcon 9 launches kept after removing Falcon 1 flights
- Missing PayloadMass values replaced with the column mean
- 26 missing LandingPad values kept, because they mean no landing pad was used
- During the lab the live API returned an SSL error (HTTP 525), so the identical IBM static snapshot of the API data was used to finish
- Used a fixed snapshot of the Wikipedia page so results are reproducible
- Helper functions cleaned dates, times, booster versions, payload mass and landing status
- Rows were only kept when the first header cell was a valid flight number
- Some cells had no hyperlink (for example certain customers), so text was read with `get_text()` as a fallback

Data Wrangling

- Load dataset_part_1.csv
- Check missing values & data types
 - Count launches by site & orbit
 - Count landing outcomes
- Map outcomes to Class (1 / 0)
 - Export dataset_part_2.csv

EDA with Data Visualization

- Flight Number vs. Payload Mass: Scatter Plot to see how experience and payload together affect landings
- Flight Number vs. Launch Site : Scatter Plot to compare success over time at each site
- Payload Mass vs. Launch Site : Scatter Plot to check payload limits and success by site
- Success Rate by Orbit Type Bar chart to compare average success across orbits
- Flight Number vs. Orbit Type : Scatter Plot to see whether experience helps in each orbit
- Payload Mass vs. Orbit Type : Scatter Plot to see how payload affects success in each orbit
- Yearly Success Trend Line chart to track the success rate from 2010 to 2020

EDA with SQL: Loaded SpaceX.csv into a SQLite database (SPACEXTBL) and ran these queries:

- Unique launch site names
- 5 records where launch sites begin with 'CCA'
- Total payload mass for NASA (CRS)
- Date of the first successful ground pad landing
- Total successful and failed mission outcomes
- Boosters that carried the maximum payload (subquery)
- 2015 drone ship landing failures by month
- Rank of landing outcomes, 2010-06-04 to 2017-03-20

Interactive Map with Folium

- Circle + Marker
- MarkerCluster
- Colored Markers
- Mouse Position
- PolyLine + DivIcon
- Each launch site with a labeled highlight (where sites are relative to the coast and equator)
- All launches grouped per site (Avoid overlapping markers at the same site)
- Green is successful landing, red is failed. Compare success rates visually by site
- Live latitude and longitude. Find coordinates of nearby features
- Lines and distances to coast, highway, railway and city. Measures how close sites are to infrastructure

Dashboard with Plotly Dash

- Dropdown (site-dropdown): drill into one site's performance
- Pie Chart: compare sites at a glance
- Range Slider payload slider: payload ranges with high success
- Scatter chart: how payload and booster affect success
- Callbacks: the analysis interactive

Predictive Analysis (Classification)

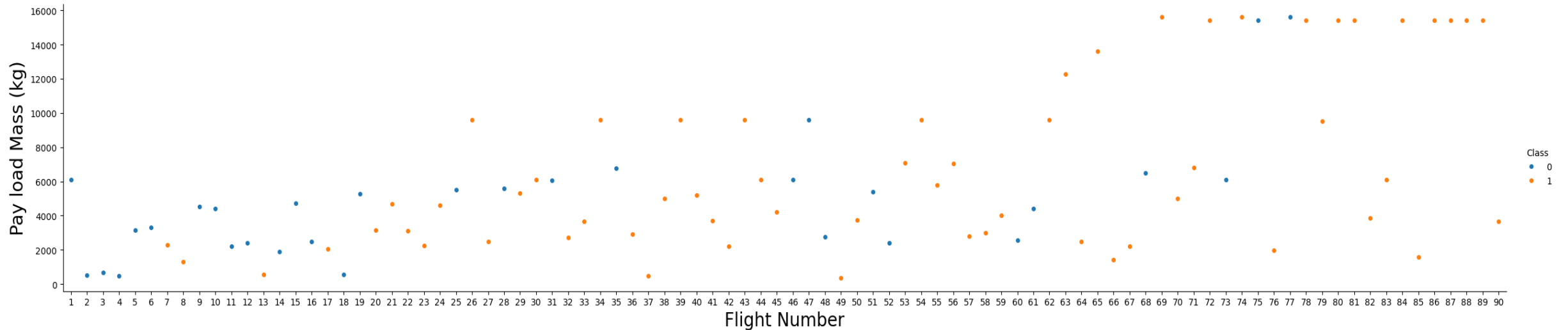
- Load features & Class label
- Standardize with StandardScaler
- Split 80/20 (18 test launches)
- GridSearchCV, 10-fold, 4 models
- Score on test data
- Compare & choose best model
- Logistic Regression
- K-Nearest Neighbors

Results

- Exploratory data analysis: visual and SQL insights on sites, orbits, payloads and time
- Interactive analytics: folium launch site maps and Plotly Dash dashboard results
- Predictive analysis: classification accuracy and confusion matrix of the best model

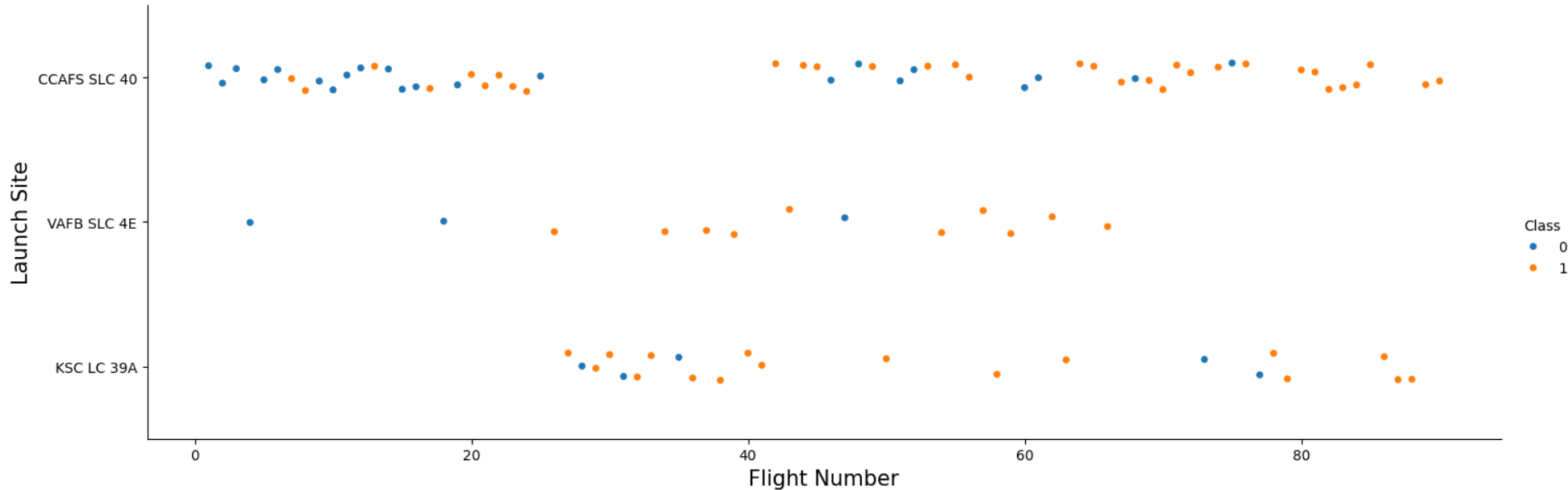
Flight Number vs. Payload MassTAB 1

- As flight number increases, the first stage is more likely to land (more orange points on the right)
- Heavy payloads above about 10,000 kg appear only in later flights, and most of those landed



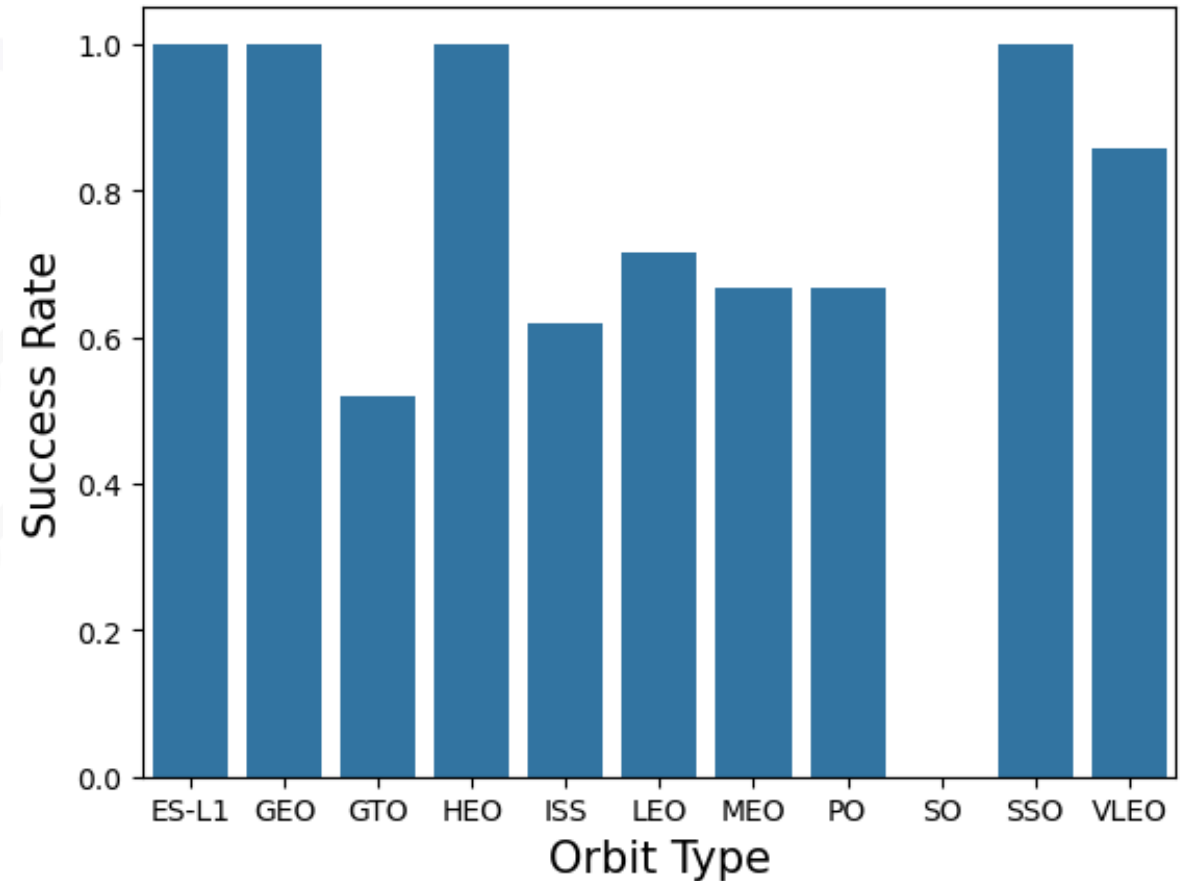
Flight Number vs. Launch Site

- CCAFS SLC 40 hosted the most launches (55); most early failures happened there
- At every site, success becomes more frequent as flight number increases, showing a clear learning curve



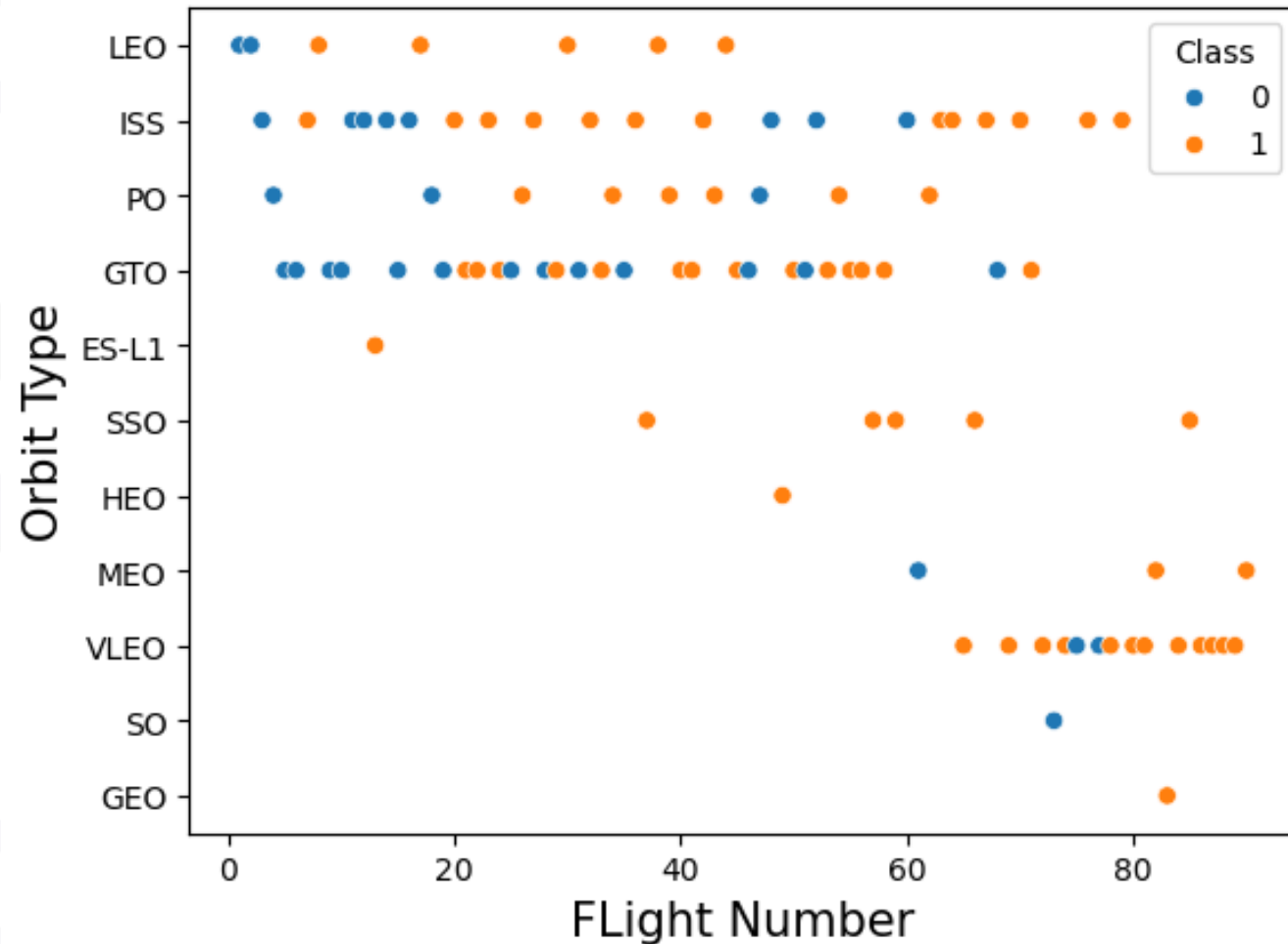
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO reached a 100% success rate
- VLEO (about 86%) and LEO (about 71%) also performed well
- GTO had the lowest success rate among common orbits, about 52%
- SO had 0% success, but from only one launch



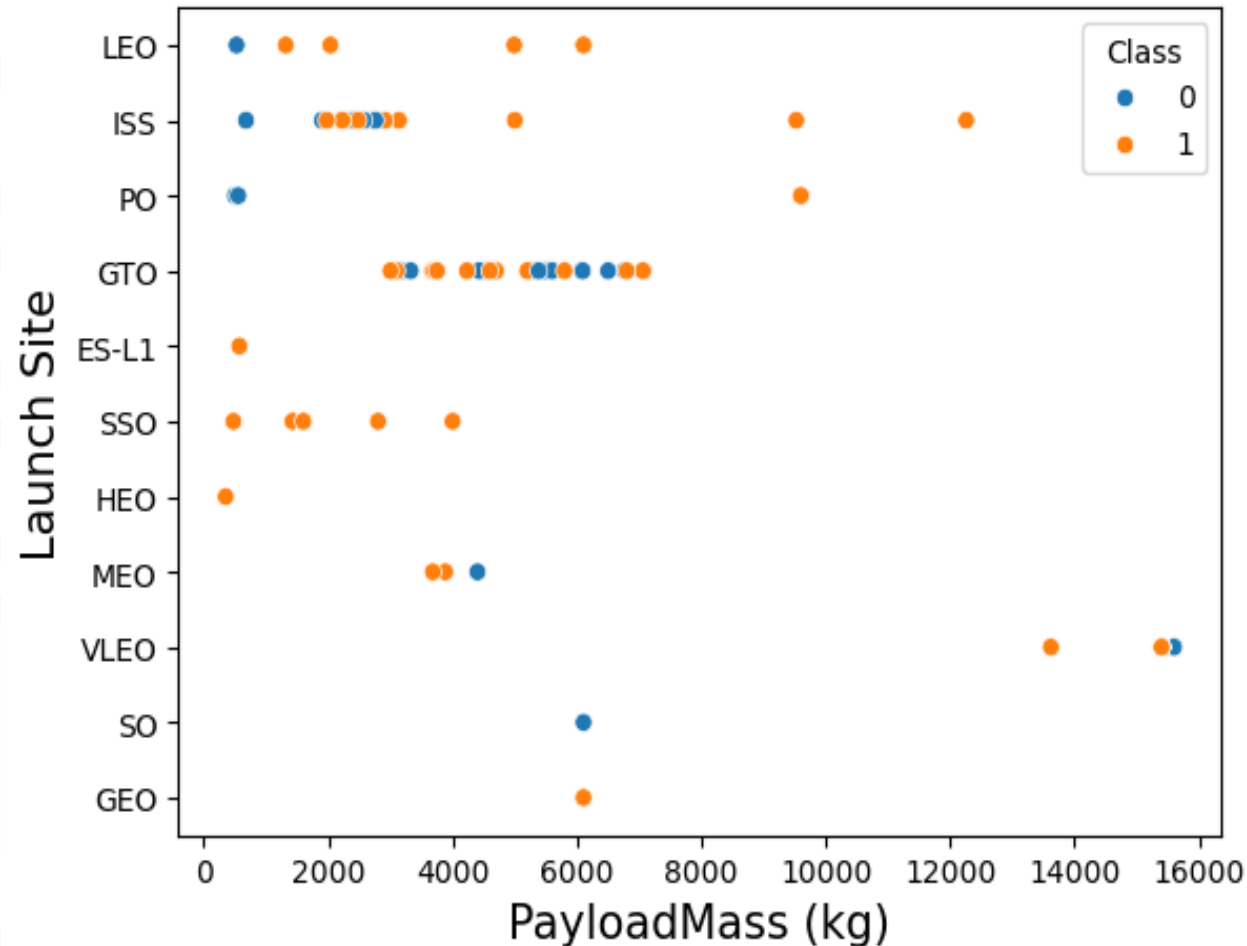
Flight Number vs. Orbit Type

- In LEO, success appears related to the number of flights
- In GTO, there is no clear relationship between flight number and success
- VLEO launches only began in later flights and were mostly successful



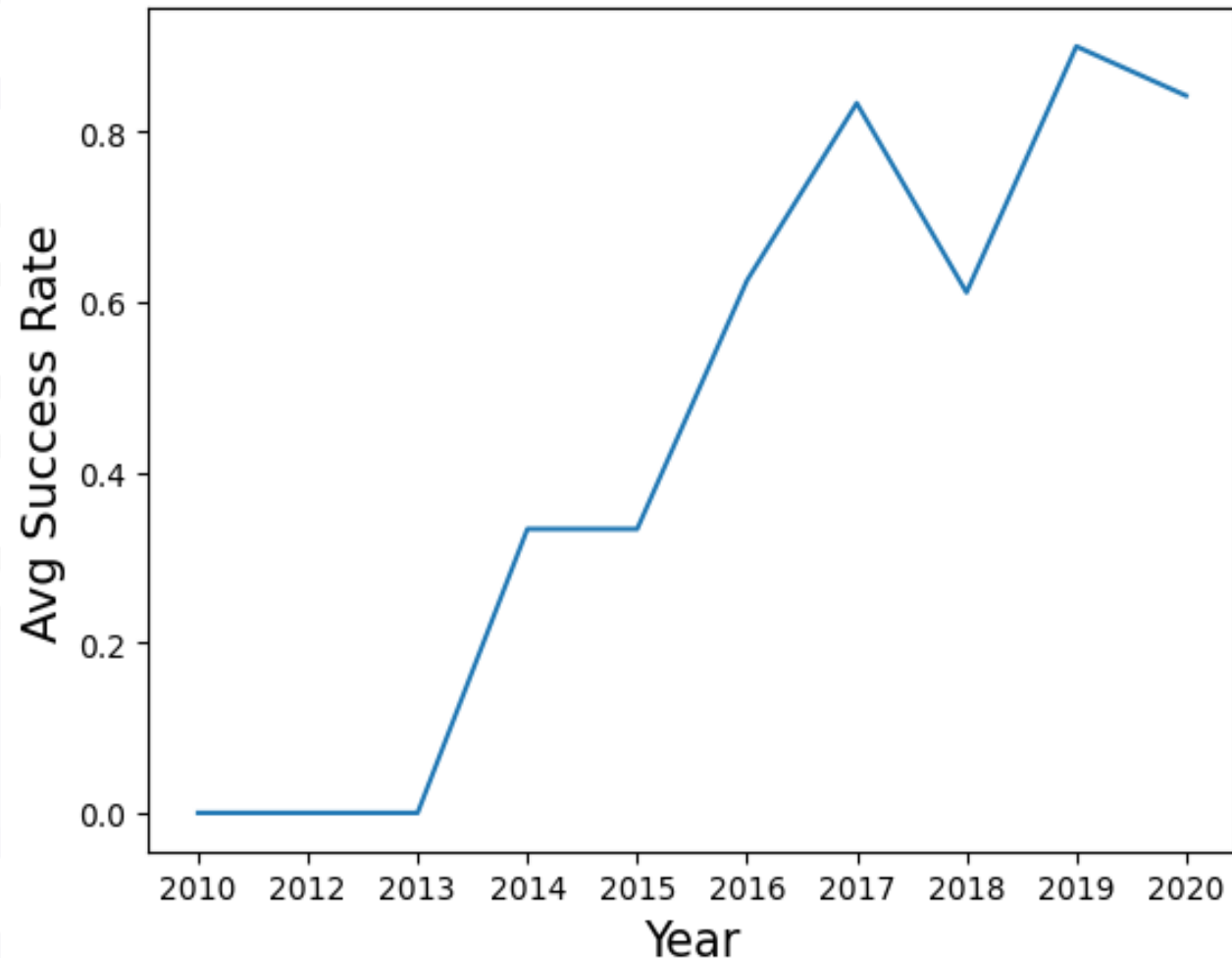
Payload vs. Orbit Type

- With heavy payloads, successful landings are more common for Polar (PO), LEO and ISS orbits
- GTO outcomes are mixed at every payload, so payload alone doesn't predict success there
- The y-axis shows orbit type



Launch Success Yearly Trend

- No successful landings from 2010 to 2013
- Success climbed steadily from 2013, reaching about 83% in 2017
- A dip in 2018 (about 61%) was followed by a peak of about 90% in 2019
- Experience is one of the strongest drivers of landing success



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- Four unique launch sites: three in Florida (Cape Canaveral and Kennedy Space Center) and one in California (Vandenberg)
 - DISTINCT removes repeated site names from the 101 launch records
 - All 5 earliest records launched from CCAFS LC-40 on v1.0 boosters to low Earth orbit
 - None achieved a booster landing: parachute recoveries failed or no landing was attempted
 - NASA's ISS resupply missions are a major, recurring customer for Falcon 9
 - Reliable, reusable boosters lower the cost of these repeated missions
 - Total payload carried for NASA's Commercial Resupply Services (45,496 kg)

• Average Payload Mass by F9 v1.1

- The early v1.1 booster carried moderate payloads
- Later Block 5 boosters carried up to 15,600 kg, more than 5 times this average

• First Successful Ground Landing Date

- Came about 5.5 years after the first Falcon 9 launch (2010-06-04)
- Marks the turning point after which landing success rose quickly

• Drone Ship Successes, Payload 4,000–6,000 kg

- All four are Full Thrust (FT) boosters
- Two (B1021.2 and B1031.2) were on their second flight, early proof that reused boosters can land again

• Total Successful and Failed Mission Outcomes

- Falcon 9 delivered its payload in about 99% of missions
- So uncertainty is the booster landing, not the mission itself
- A trailing space split one 'Success' into its own group, a data quality issue worth cleaning
- 100 success and 1 failure

• Boosters That Carried the Maximum Payload

- All 12 are Block 5 boosters carrying the maximum payload of 15,600 kg
- Many were on their 4th to 7th flight (e.g., B1049.7), so the heaviest loads fly on heavily reused boosters

• 2015 Launch Records

- Both 2015 drone ship failures used v1.1 boosters from CCAFS LC-40
- Drone ship landings succeeded only after the move to Full Thrust boosters

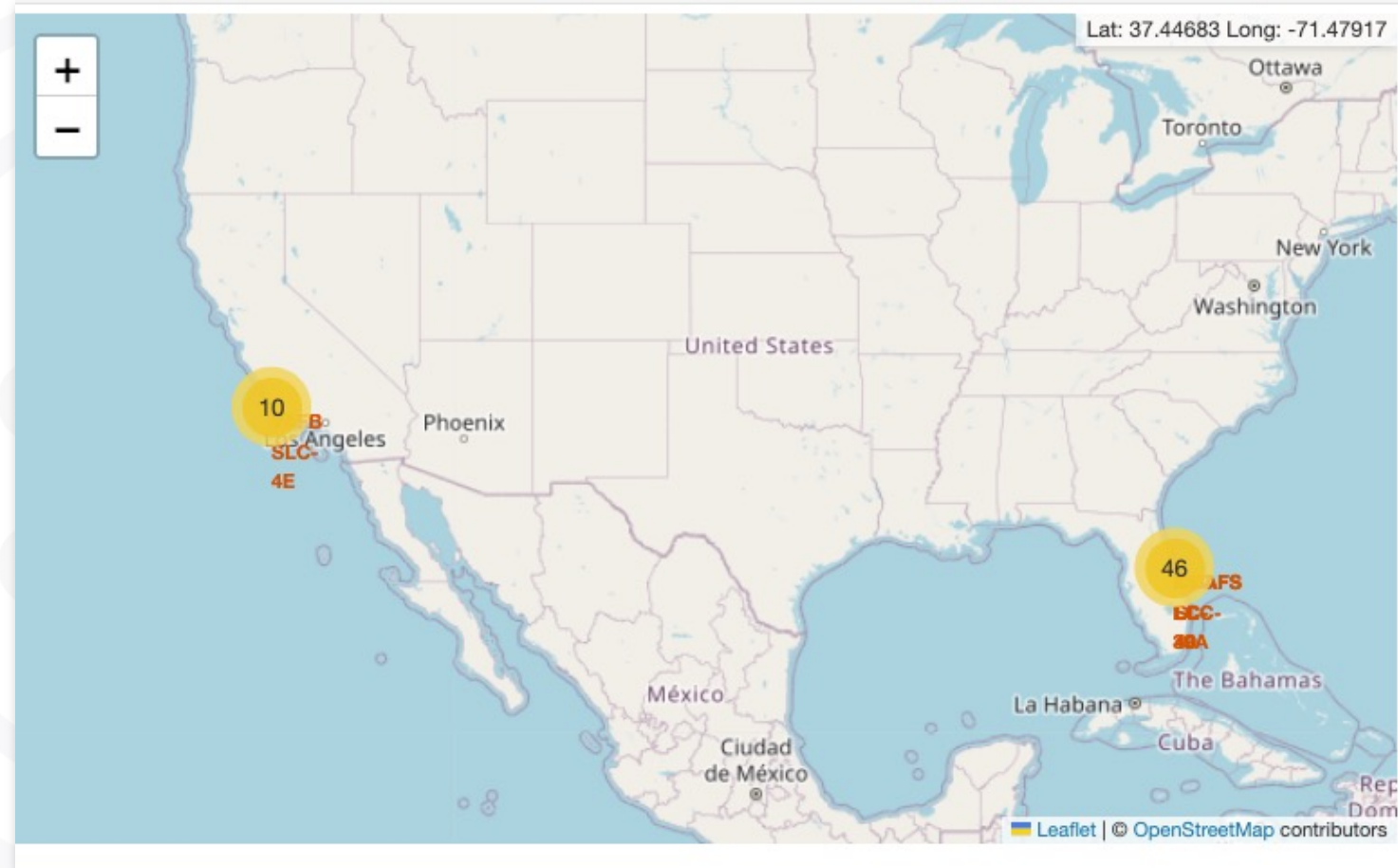


Rank Landing Outcomes (2010 to 2017)

- 'No attempt' was the most common outcome (10)
- Drone ship landings split evenly: 5 successes and 5 failures
- Ground pad landings went 3 for 3

All Launch Sites on a Global Map

- Four launch sites: three on Florida's east coast and one on California's coast
- Marker clusters show 46 launches from Florida and 10 from California
- Every site is on a coast, so rockets fly over water, away from people
- Sites are as far south as possible in the U.S; closer to the equator, Earth's rotation gives rockets a speed boost



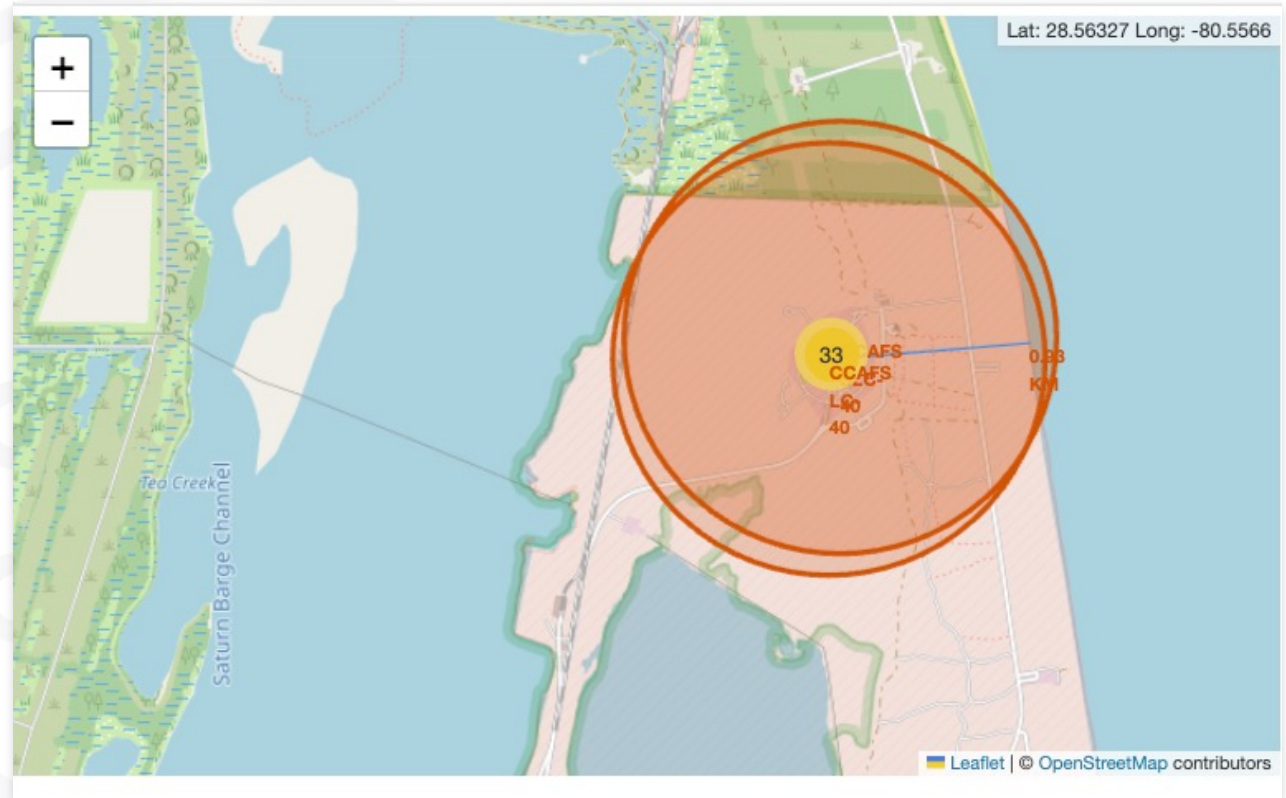
Launch Outcomes by Site

- Zoomed in on Cape Canaveral: clusters hold 26 launches at CCAFS LC-40 and 7 at CCAFS SLC-40
- Clicking a cluster expands it into green (success) and red (failure) markers
- CCAFS LC-40 landed only 7 of 26 (27%), since it hosted most early test flights
- KSC LC-39A had the best record: 10 of 13 (77%)



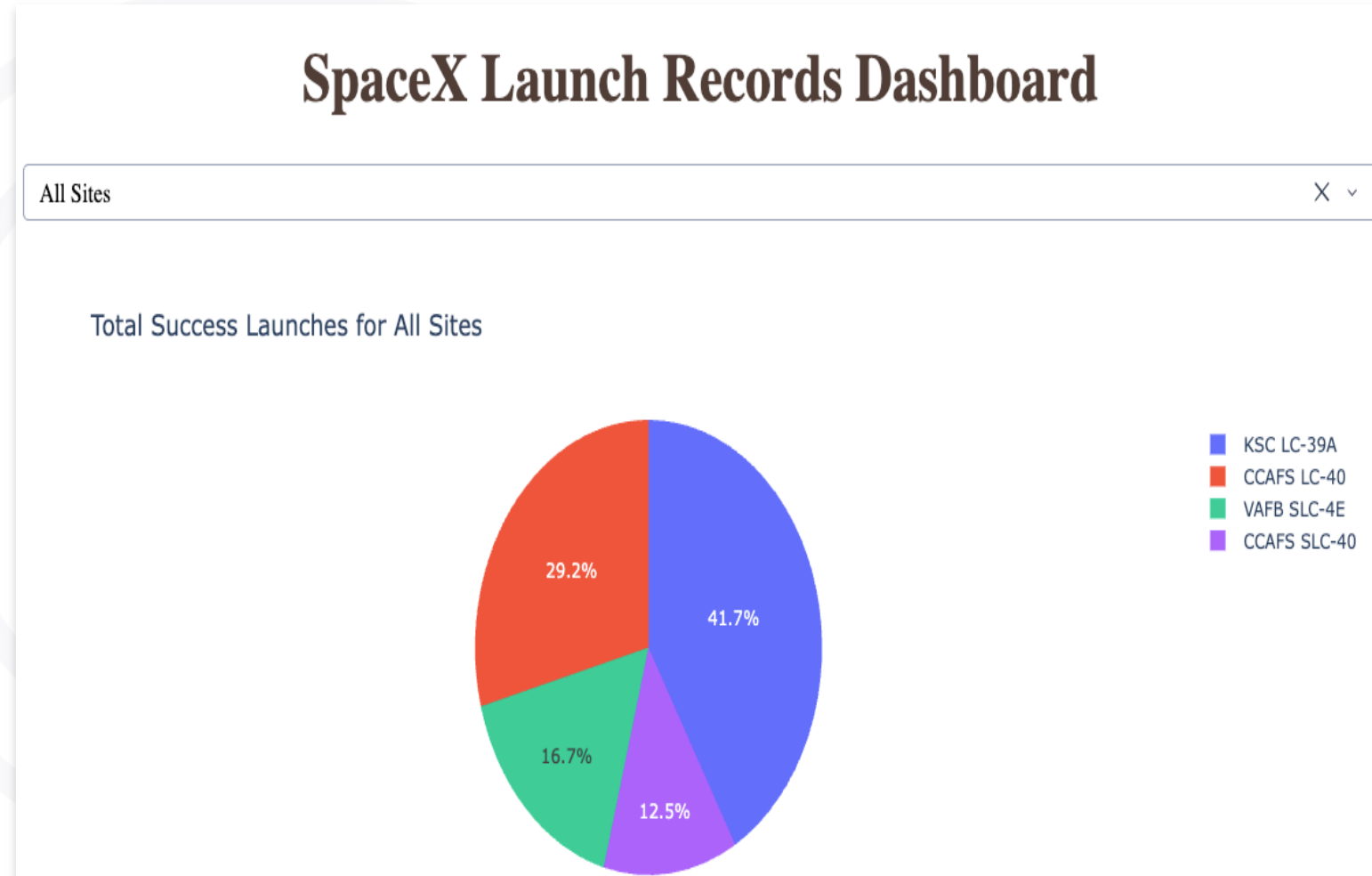
Launch Site Proximity to Infrastructure

- Sites sit within about 1 km of highways, railways and the coast for easy transport of heavy parts
- They stay more than 20 km from cities for public safety



Flight Number vs. Orbit Type

- KSC LC-39A accounts for the largest share of successful launches (41.7%)
- CCAFS LC-40 follows at 29.2%, then VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%)
- Share of successes isn't the same as success rate: CCAFS LC-40 had many launches but a low rate



Launch Site with the Highest Success Rate: KSC LC-39A

- KSC LC-39A landed 10 of 13 boosters, the highest success rate of any site
- It also hosted mostly later flights, after landing technology matured

Payload vs. Launch Outcome for All Sites

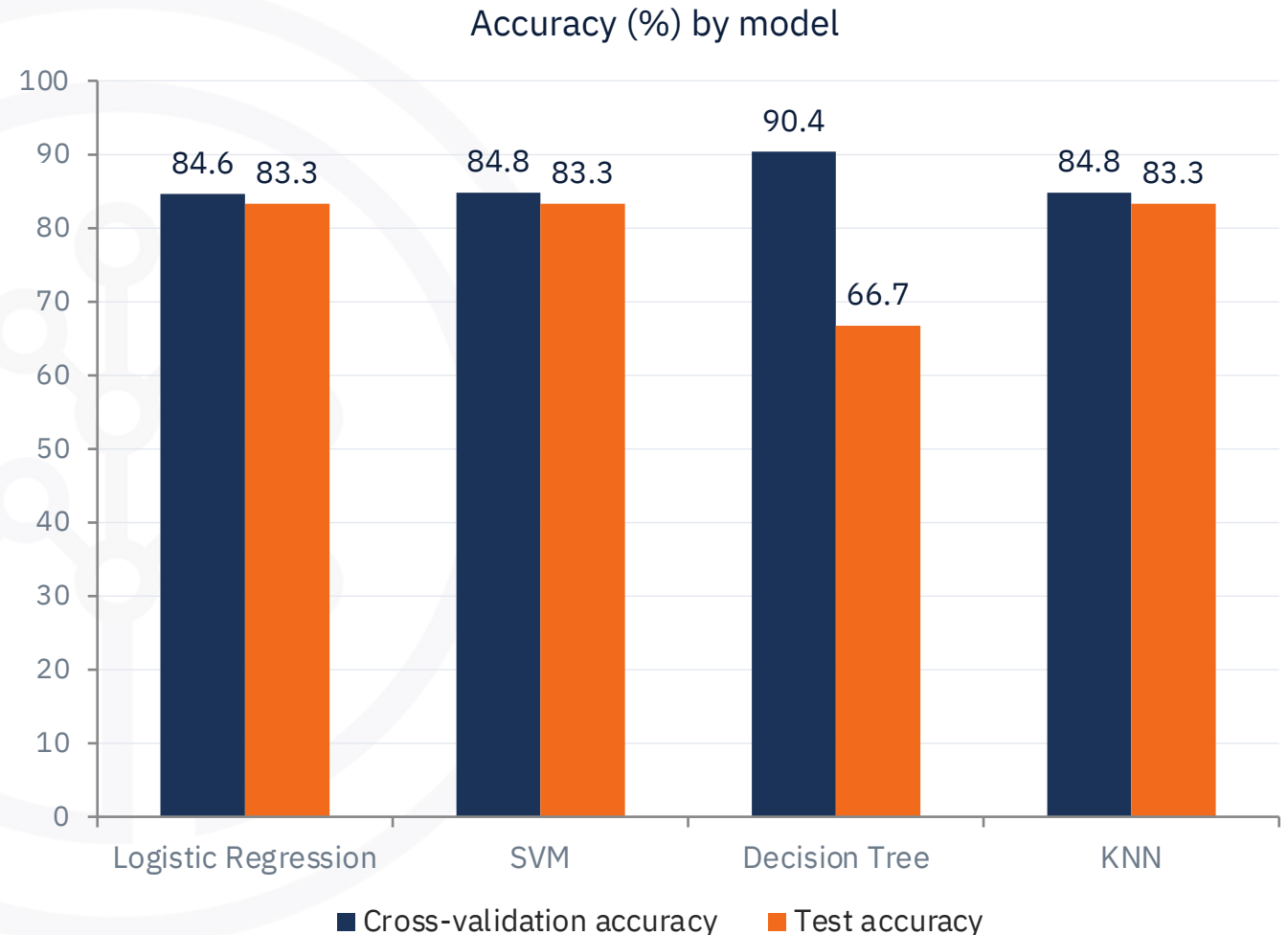
Payloads of 2,000–4,000 kg had the highest success rate (62%)

- No launches between 6,000 and 8,000 kg landed
- FT boosters succeeded most often (67%), while v1.1 landed only 1 of 15 and v1.0 none
- The booster version matters more than payload alone



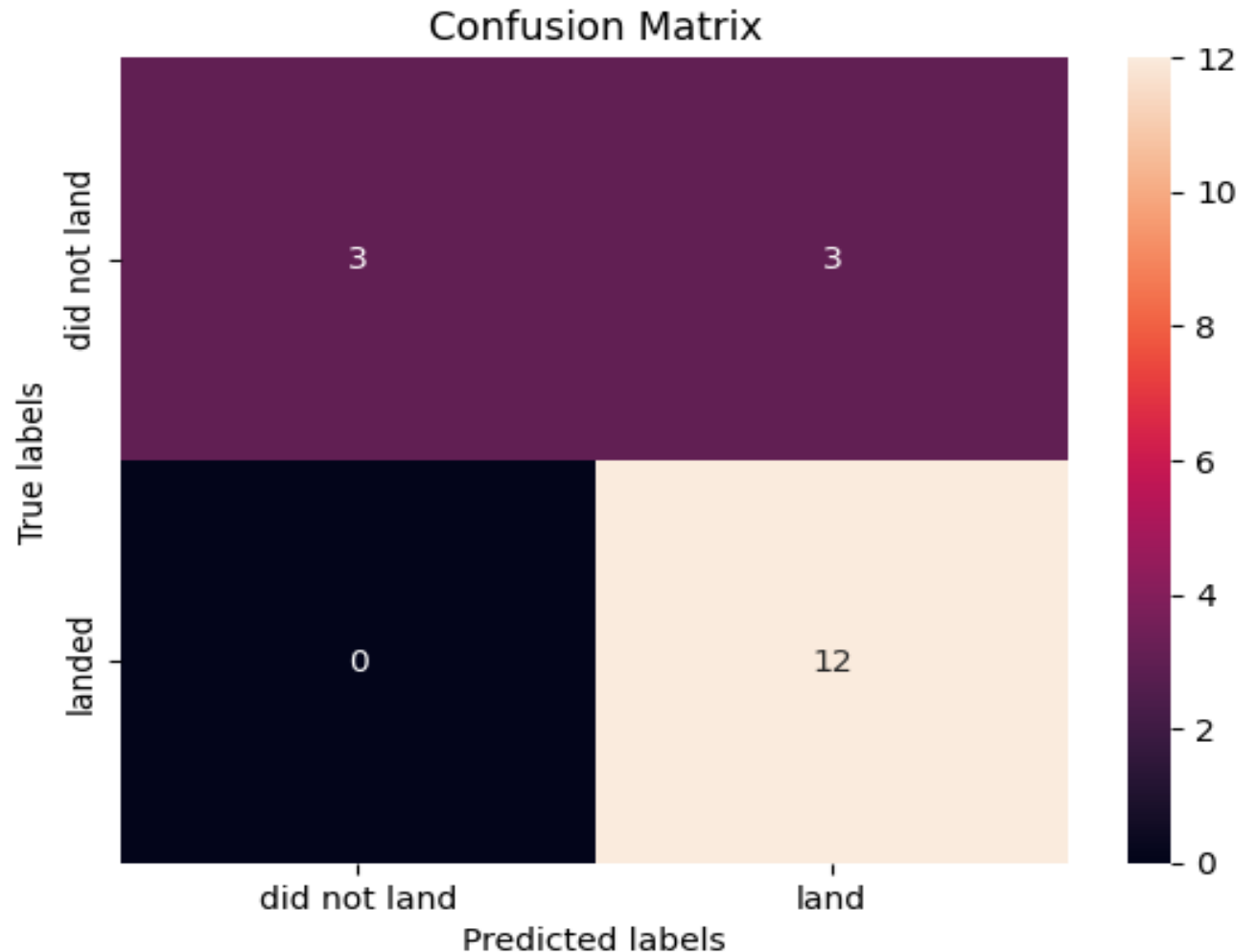
Classification Accuracy

- Logistic Regression, SVM and KNN tie at 83.3% test accuracy (15 of 18 correct)
- Recommend Logistic Regression: same accuracy, and easiest to explain
- Decision Tree scored highest in cross-validation (90.4%) but lowest on test data (66.7%), a sign of overfitting



Confusion Matrix

- 12 true positives: landings correctly predicted
- 3 true negatives: failures correctly predicted
- 3 false positives: failures predicted as landings
- 0 false negatives: no landing was missed
- Logistic Regression, SVM and KNN produced this same matrix; the main weakness is false positives



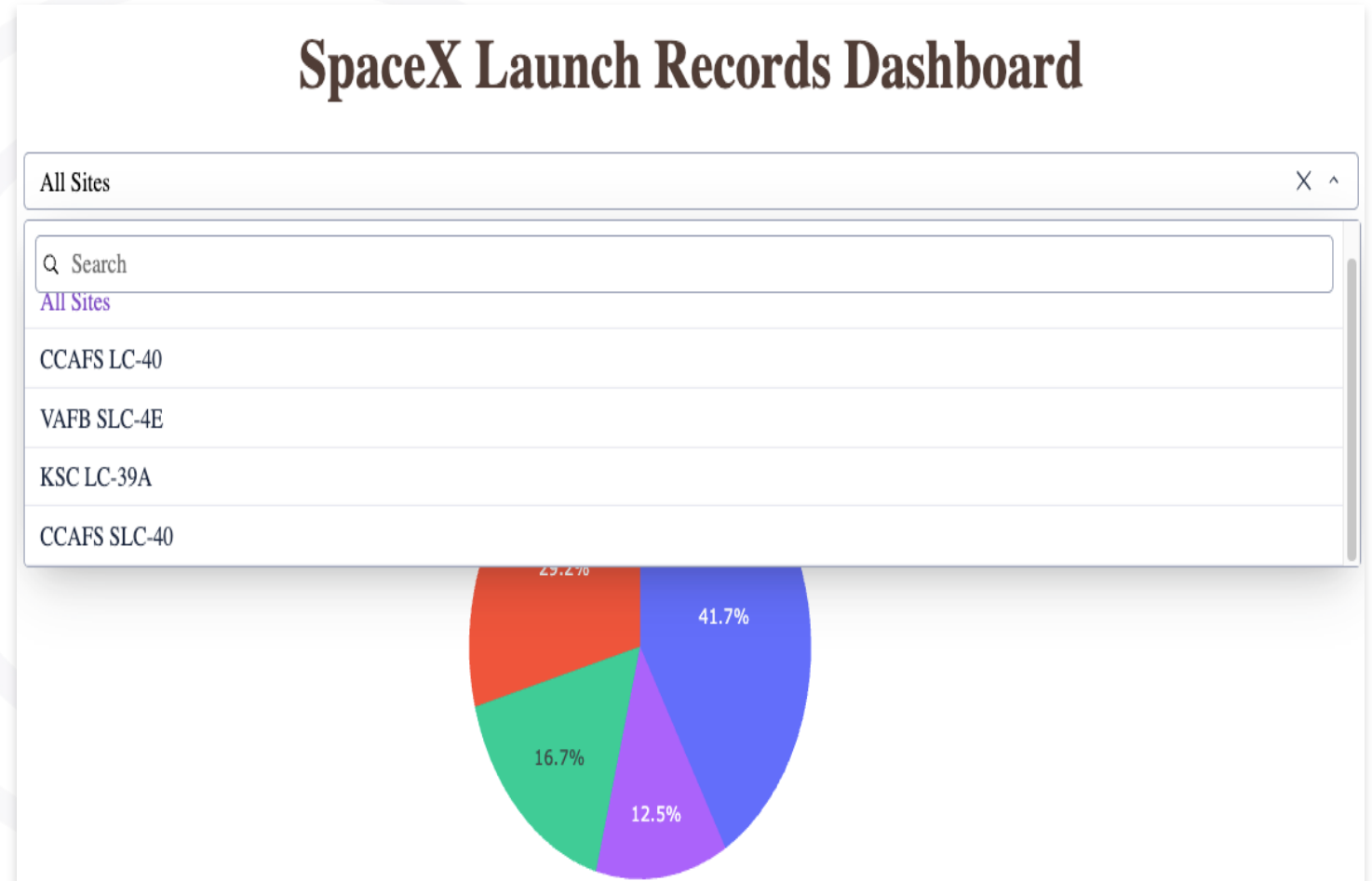
Conclusions

- Landing success rose from 0% (2010–2013) to about 90% in 2019 as flight count grew
- KSC LC-39A had the highest success rate (77%); all sites are coastal and near transport links
- ES-L1, GEO, HEO and SSO had 100% success; GTO was hardest. 2,000–4,000 kg payloads did best
- ES-L1, GEO, HEO and SSO had 100% success; GTO was hardest. 2,000–4,000 kg payloads did best

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- The Decision Tree led cross-validation (90.4%) but matched a naive 'always predict landing' guess on test data (66.7%). With only 18 test launches, each prediction moves accuracy 5.6 points, so repeated cross-validation would give a better comparison.
 - ES-L1, GEO and HEO each show 100% success from a single launch. Rates based on so few launches should be treated with caution in decisions.
 - They never missed a landing but incorrectly predicted landings for 3 of 6 failures. For a competitor bidding against SpaceX, that means underestimating SpaceX's costs; tuning for fewer false positives would add business value.
 - The live SpaceX API failed with an SSL error (HTTP 525) and Wikipedia's table format broke the scraper. Static snapshots and parsing kept the analysis reproducible.

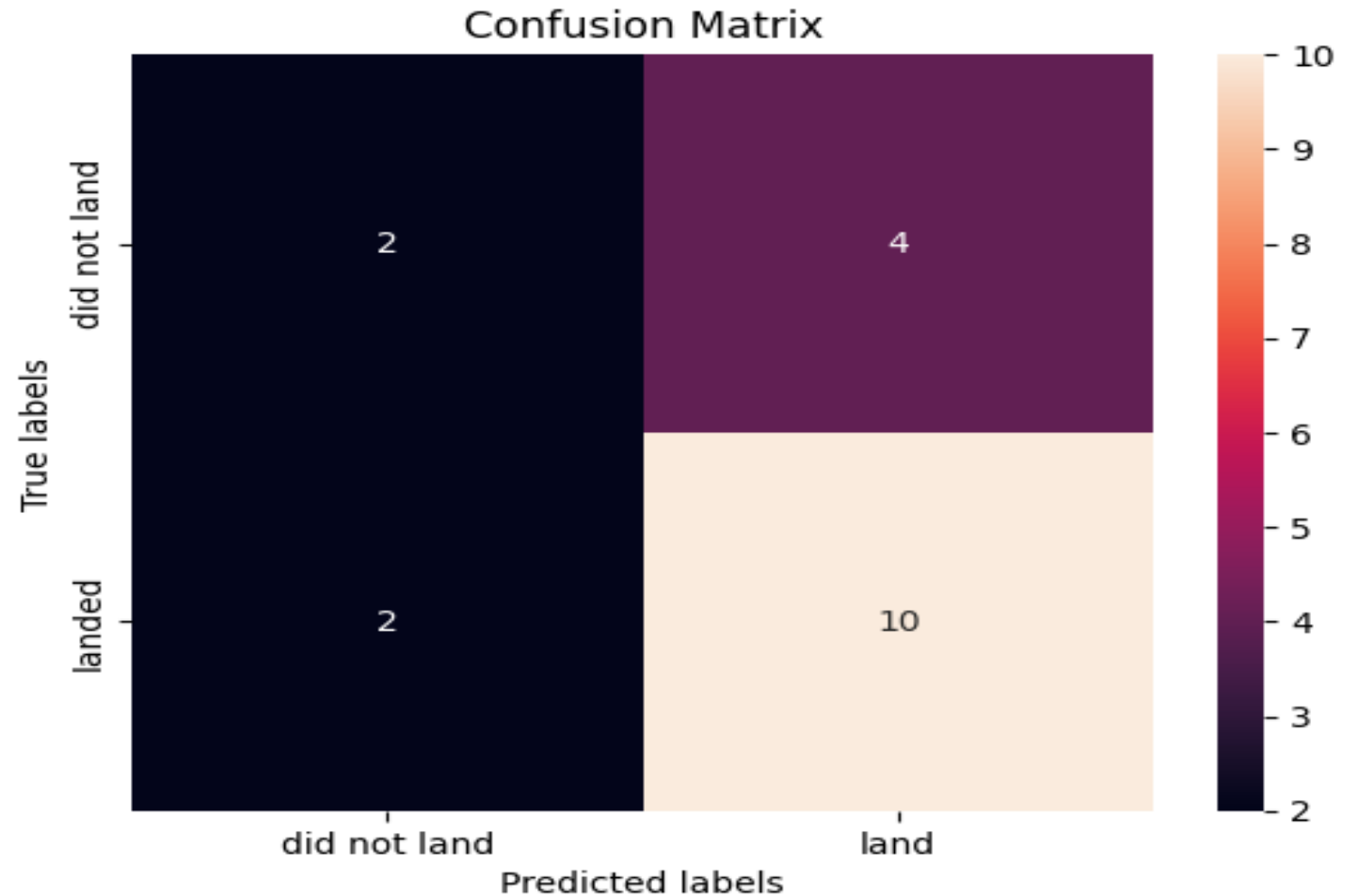
Appendix: Dashboard Launch Site Dropdown

- All Sites (default)
- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40
- The dropdown is searchable and updates both the pie and scatter charts



Appendix: Decision Tree Confusion Matrix

- 10 true positives and 2 true negatives (12 of 18 correct)
- 4 false positives and 2 false negatives, the only model that missed real landings
- max_depth = 16 on just 72 training launches let the tree memorize noise



Results



- Landing success increased from 0% (2010–2013) to about 90% by 2019
- KSC LC-39A has the highest success rate of any site
- ES-L1, GEO, HEO and SSO orbits reached 100% success; GTO is the hardest (about 52%)
- Logistic Regression, SVM and KNN predict landings with 83.3% test accuracy
- Decision Tree overfit: best in cross-validation (90.4%) but worst on test data (66.7%)

Appendix: SQL Query Corrections



1. Launch site names: returned every row (101) instead of unique sites
2. First ground pad landing: single quotes made 'Landing_Outcome' a text literal, returning None
3. First ground pad landing: query cell was missing
4. 2015 records: a missing comma hid the Booster_Version column